

Quantifying Uncertainty in Aya-Expanse Language Models for Persian with an Additional Head

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Abstract—The Aya-Expanse family of language models represents a significant advancement in small language models, particularly in their ability to communicate effectively in low-resource languages while minimizing issues related to language mixing challenges often encountered by smaller-scale models. However, despite these strengths, Aya-Expanse models exhibit a critical deficiency: they struggle with uncertainty in their responses, which complicates the reliability of the information they provide. This paper addresses this limitation by proposing the development of an uncertainty head specifically designed for the Aya-Expanse models. By implementing this uncertainty head, we aim to quantify the model’s confidence in its responses using its internal features, thereby enhancing its overall reliability and utility. Our findings demonstrate that this approach not only improves the interpretability of the model’s outputs but also fosters greater trust in its applications across various linguistic contexts.

Index Terms—Small Language Models, Aya-Expanse, Uncertainty Quantification, Low Resource Language

I. INTRODUCTION

In the rapidly evolving field of artificial intelligence, the development of language models has highlighted a significant issue: the data imbalance across different languages. While languages like English and Chinese dominate the available training datasets, many low-resource languages, such as Persian, only comprise a small fraction of this data. [1] This imbalance leads to various challenges for small language models, including language mixing and an inability to accurately learn the grammar and nuances of underrepresented languages. To address this disparity, one promising approach is the use of synthetic data generated by other language models. However, an over-reliance on synthetic data can trigger a phenomenon known as model collapse [2] where the model’s performance deteriorates due to a lack of diverse training inputs. Recent research from the C4AI team has shown that it is possible to mitigate model collapse by employing a pool of generators and a strategy called data arbitrage [3] which led to the creation of a new model called Aya-Expanse [4].

Aya-Expanse, despite its success, faces significant challenges related to the uncertainty of its responses, often producing different and sometimes contradictory answers to the same query upon multiple runs. This variability likely stems from its reliance on a data arbitrage strategy, which may complicate consistency compared to other language models trained on natural data. As a result, Aya-Expanse struggles with reliability, raising concerns about the correctness of its responses.

In scenarios where a language model exhibits significant uncertainty about its outputs, it becomes crucial to quantify that uncertainty effectively in order to assess the reliability of its responses. When a model is identified as uncertain about its answer, it can be reasonably concluded that the generated answer is produced due to model hallucination not its knowledge; therefore, we don’t rely on the generated answer. Conversely, if we can ascertain that the model is confident in its response, we can place greater trust in the information it provides on that response. This distinction is vital, as it not only influences how users interpret and utilize the model’s outputs but also informs the development of strategies to enhance the model’s performance and reliability.

One method to assess a model’s certainty involves directly asking it about its confidence, but this requires a level of self-awareness that smaller models like Aya-Expanse lack. An alternative approach is to run multiple iterations of the model and compare the responses, which can offer insights into uncertainty but is often costly and time-consuming. These methods are typically used with black box language models, where only the final output is accessible. In contrast, when working with a white box language model like Aya-Expanse, we can utilize its internal features to estimate uncertainty more effectively by adding an additional head that provides a numerical representation of the model’s uncertainty, separate

from its generated sequence of tokens.¹

In this work, we aim to develop the aforementioned uncertainty head for the Aya-Expanse family of language models. To summarize, our contributions are as follows:

- 1) We raise awareness about the uncertainty in Aya-Expanse and develop an additional head to quantify its uncertainty for each response that it may produce given a query.²
- 2) We propose a method to adapt the uncertainty head developed for another language model and language into a new language model and language.

II. BACKGROUND

A. Uncertainty

As language models are increasingly deployed in high-stakes environments, the ability to quantify uncertainty becomes paramount. [5] This is particularly true when considering that uncertainty can inform users about the model’s confidence in its predictions, thereby guiding decision-making processes. Understanding uncertainty allows researchers and practitioners to identify potential weaknesses in a model’s training data, which can manifest as either epistemic uncertainty—arising from insufficient training data on specific queries—or aleatoric uncertainty, which stems from inherent variability or disagreement within the training data.

One significant implication of estimating uncertainty is its role in detecting hallucinations within language models. Hallucination refers to instances where models generate outputs that are factually incorrect or nonsensical. By evaluating a model’s uncertainty, researchers can pinpoint when a model is likely to produce unreliable information, enabling them to take corrective measures or apply additional scrutiny to the generated content. This capability is particularly relevant in applications where factual accuracy is critical, such as medical diagnosis or legal advice. [6]

Various methodologies have been proposed for quantifying uncertainty in language models, as classified by Zhiqiu Xia et al. [7] These approaches offer diverse strategies for assessing model confidence. Verbal methods involve directly querying the model about its own uncertainty, providing a straightforward way to gauge confidence levels. Latent information methods leverage internal features of the language model, extracting insights from the hidden layers that may indicate how certain or uncertain a model is about its predictions. Consistency-based methods enhance reliability by generating multiple samples from the model and evaluating their consistency, which serves as a proxy for uncertainty. Finally, semantic clustering methods involve generating multiple outputs and clustering the resulting tokens to estimate uncertainty, allowing for a more nuanced understanding of the model’s performance across different contexts.

Moreover, recent research highlights the potential of using uncertainty as an alternative to traditional process reward

models. [8] In this context, uncertainty has been employed to assess the validity of atomic claims within a Chain of Thought framework. By determining whether a claim is accurate based on the associated uncertainty, researchers can refine the model’s reasoning capabilities and improve overall performance. This innovative application underscores the versatility of uncertainty estimation and its potential to enhance the robustness of language models across various tasks.

B. Aya-Expanse

The Aya-Expanse language model represents a significant advancement in the realm of low-resource languages, as discussed in the introduction section. By leveraging synthetic data, Aya-Expanse has effectively learned to understand and generate underrepresented languages. Its training pipeline is noteworthy for its inclusion of fine-tuning on synthetic data, iterative preference training [1], and model merging. [9] This innovative approach allows Aya-Expanse to harness diverse data sources, but it also raises questions about the model’s reliability and the uncertainty inherent in its outputs.

A simple experiment has been conducted to demonstrate the uncertainty present in both the Aya-Expanse and Gaokerena models. In this experiment, Aya-Expanse, Gaokerena-V, Gaokerena-R, and Med-Gemma are utilized to tackle the Iranian Basic Medical Science Entrance Exam held in September 2023³. Each model was run five times on the same set of questions. The results revealed that Med-Gemma consistently selected the same option for all questions, while the other models exhibited variability, choosing the same option only for some questions. These findings are summarized in Table I. This experiment underscores the level of uncertainty associated with Aya-Expanse and its variants, highlighting the challenges these models face in providing reliable and consistent outputs.⁴

Efforts have been made to enhance Aya-Expanse’s knowledge in specific languages and domains, such as the Gaokerena initiative [11] [12], which has focused on improving the model’s medical knowledge in Persian. As you can observe in Table I, although Gaokerena-V has gained more Persian medical knowledge by getting trained on 57M Persian medical tokens, it has become more uncertain than Aya-Expanse. This fact warns us about the vulnerability of Aya-Expanse when it comes to being trained on new tokens, as there is a danger of hallucination during the training process. While these targeted interventions aim to bolster the model’s expertise, they too encounter the challenge of uncertainty, mirroring the issues faced by the baseline model. As researchers continue to refine Aya-Expanse, addressing this uncertainty will be crucial for ensuring that it can reliably serve its intended purpose in low-resource language contexts.

One possible explanation for the uncertainty phenomenon observed in Aya-Expanse could be attributed to the varied teacher models that have been employed to generate synthetic

¹All of our work is open-source and available at github.com/Mehrdadghassabi/Gaokerena-uh

²available at huggingface.co/gaokerena/gaokerena_uncertainty_head

³Available at huggingface.co/datasets/gaokerena/KOPP

⁴we have used chain of thought [10] prompting since it helps to unveil uncertainty more clearly

data. Each of these large language models brings its own unique perspectives and biases, resulting in a rich yet inconsistent dataset for Aya-Expanse’s training. The amalgamation of these differing ideas may contribute to the uncertainty that users experience when interacting with the model.

TABLE I
COMPARISON OF MODEL CONSISTENCY ON IBMSEE

	gaokereana-V	aya-expanse	medgemma
Accuracy	38.69	34.52	37.5
Number of Single Option Taken	10	44	168
Number of All Option Taken	10	7	0
Entropy	1.62	1.21	0
Prompt	COT	COT	COT
Number of Parameters	8 billion	8 billion	4 billion

III. METHODOLOGY

A. Uncertainty Head Framework

We build our approach upon the LLM Uncertainty Head framework [6]. The framework augments a pretrained language model with an additional lightweight classification head designed to estimate token-level uncertainty in generated responses. Rather than modifying the backbone architecture, the method introduces a separate prediction head trained to identify tokens associated with hallucinated claims.

Given an input prompt and its generated response, atomic claims are first extracted from the response. These claims are then aligned with their corresponding token spans in the original text. The uncertainty head is trained as a token-level classifier: tokens belonging to hallucinated claims receive label 1, tokens belonging to supported claims receive label 0, and non-claim tokens are excluded from training by assigning a masking value of -100 in the loss computation. This design enables efficient supervision at the token level while preserving the pretrained knowledge of the base model.

In this work we have extended this framework originally introduced for the mistral-7b model [13] and English, to adapt it for Persian within the Aya-Expanse model family.

B. Adaptation to Aya-Expanse

We apply the uncertainty head training procedure to Aya-Expanse, using the same token-level supervision paradigm introduced in the original work. Since the original dataset is constructed in English, a direct application to Persian requires a carefully designed translation and alignment pipeline to ensure structural and semantic consistency between sentences, claims, and token spans. We therefore construct a Persian training dataset that preserves claim boundaries and token-level alignment.

C. Persian Data Construction Pipeline

To enable uncertainty head training in Persian, we translate the original English dataset and reconstruct sentence-level and token-level alignments. The pipeline is designed to maintain

semantic equivalence while ensuring compatibility with the Persian tokenizer of Aya-Expanse. The steps are as follows:

- 1) Translation of Questions and Responses: English questions and generated responses are translated into Persian using the DeepSeek-V3 model [14]. This step produces fluent Persian text while preserving the structure and meaning of the original English output.
 - 2) Sentence-Level Alignment: Since claims are extracted at the English sentence level, sentence-level correspondence between English and Persian must be established. The translated Persian responses are first segmented into individual sentences. For each English sentence associated with extracted claims, we identify its aligned Persian sentence using two independent methods:
 - a) LaBSE Similarity Matching [15]: The English sentence and all Persian candidate sentences are embedded using LaBSE. The Persian sentence with the highest cosine similarity is selected.
 - b) Qwen3-Reranker Matching: The English sentence is treated as a query and the Persian candidate sentences as documents. The Qwen3-reranker-0.6B [16] model ranks the candidates according to translation equivalence. The highest-scoring candidate is selected.
- A sentence pair is retained only if both LaBSE and Qwen3-reranker agree. Otherwise, the English sentence and its claims are removed from the dataset.
- 3) Claim Translation: Once sentence-level alignment is established, the extracted English claims are translated into Persian. To ensure terminological and structural consistency, we provide the translated Persian sentence (from which the claim originates) to DeepSeek-V3 and instruct it to translate the claim in a manner consistent with the Persian sentence.
 - 4) Token-Level Alignment: Following claim translation, Persian claim tokens are aligned with the tokens of the corresponding Persian sentence, following the methodology described in the original uncertainty head framework. Tokens corresponding to claim spans are assigned the appropriate label, while all other tokens are masked with -100 and excluded from loss computation.

D. Training Procedure

Using the constructed Persian dataset, we train the uncertainty head on top of Aya-Expanse-8B using token-level supervision. The objective is a cross-entropy loss computed only over labeled tokens. The backbone model parameters are frozen, while the uncertainty head parameters are optimized during training. Detailed training configurations, including optimizer settings and hyperparameters, are provided in Table II.

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TABLE II
TRAINING HYPERPARAMETERS

Number of epochs	10
Learning rate	0.0001
Warmup ratio	0.05
Weight decay	0.1
Batch size	32
Number of gradient accumulation steps	1
Maximum gradient norm	1
Uncertainty head dimension	768
Uncertainty head layers	2
Uncertainty heads	8
Uncertainty head drop out	0.1

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